C Programming & Data Structures Lab Assignments

Assignment Problem Statements & GitHub Submission Guidelines

Total Assignments: 10

Language: C
Submission Platform: GitHub
Evaluation: Code + Documentation + GitHub Repository

Assignment List

Assignment	Title	Problem Statement	Concepts Covered	Expected Deliverables
01	Student Marks Analyzer	Develop a menu-driven application to store marks of **N students**. The program should display marks, calculate highest, lowest, total, average, search for a student's marks, and sort marks in ascending order.	Arrays, Loops, Functions, Searching, Sorting	Source Code, README, Screenshot
02	String Utility Toolkit	Develop a menu-driven program to perform string operations such as finding length, reversing a string, checking palindrome, counting vowels, digits, spaces, and displaying character frequency.	Strings, Arrays, Loops, Functions	Source Code, README, Screenshot

03	Pointer Playground	Write a program demonstrating pointer concepts including displaying memory addresses, dereferencing pointers, swapping two numbers using pointers, traversing an array using pointers, and pointer arithmetic.	Pointers, Arrays, Memory Management	Source Code, README, Screenshot
04	Student Record Management System	Create a Student Management System using structures. The program should add, display, search, update, and delete student records using arrays of structures.	Structures, Arrays, Strings, Functions	Source Code, README, Screenshot
05	Dynamic Array Management	Create a program that dynamically allocates memory for storing student marks using `malloc()`. Display marks, calculate statistics, and release memory using `free()`.	Dynamic Memory Allocation, Pointers, Arrays	Source Code, README, Screenshot
06	Inventory Management System	Design an inventory system to manage products. Store Product ID, Name, Price, and Quantity. Implement add, search, update, sort by price, and calculate total inventory value.	Structures, Arrays, Sorting, Searching	Source Code, README, Screenshot
07	Hospital Patient Management	Develop a Patient Management System to register patients, display records, search by Patient ID, update patient information, and delete records using structures.	Structures, Strings, Functions, Arrays	Source Code, README, Screenshot

08	Employee Payroll System	Create an Employee Payroll System that accepts employee details and calculates HRA, DA, Tax, Gross Salary, and Net Salary. Display salary reports and sort employees based on salary.	Structures, Arrays, Functions, Conditional Statements	Source Code, README, Screenshot
09	Dynamic Array Operations	Implement insertion, deletion, updating, searching, and displaying elements in a dynamic array. Compare Linear Search and Binary Search performance after sorting.	Arrays, Searching, Dynamic Memory, Algorithm Analysis	Source Code, README, Screenshot
10	LeetCode Array Challenge	Solve an array-based challenge to find largest, second largest, smallest, second smallest, reverse the array, remove duplicates, count frequency, perform searching, and sort the array. Analyze time complexity for each operation.	Arrays, Algorithm Analysis, Searching, Sorting	Source Code, README, Complexity Analysis, Screenshot

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GitHub Submission Instructions

Step	Instructions
1	Create a **public GitHub repository** named **`C-Programming-Lab`**.
2	Create a separate folder for each assignment using the naming convention **`Assignment-01`**, **`Assignment-02`**, ..., **`Assignment-10`**.
3	Each assignment folder must contain the following files: `main.c`, `README.md`, and at least one output screenshot (`output.png` or `output.jpg`).
4	Write clean, modular, and well-commented C code using meaningful variable and function names.

5	The `README.md` file should include: Objective, Problem Statement, Algorithm, Time Complexity, Space Complexity, Sample Input, Sample Output, and Learning Outcome.
6	Test your program with multiple input cases before submission. Ensure the program compiles without errors or warnings.
7	Commit your work regularly using meaningful commit messages (e.g., `Completed Assignment 03 - Pointer Playground`).
8	Push all assignments to the GitHub repository before the submission deadline.
9	Submit only the GitHub repository link through the LMS/Google Form/Classroom as instructed by the faculty.
10	Late submissions or repositories without proper folder structure/documentation may attract evaluation penalties.

Repository Structure

Folder	Contents
Assignment-01	`main.c`, `README.md`, `output.png`
Assignment-02	`main.c`, `README.md`, `output.png`
Assignment-03	`main.c`, `README.md`, `output.png`
Assignment-04	`main.c`, `README.md`, `output.png`
Assignment-05	`main.c`, `README.md`, `output.png`
Assignment-06	`main.c`, `README.md`, `output.png`
Assignment-07	`main.c`, `README.md`, `output.png`
Assignment-08	`main.c`, `README.md`, `output.png`
Assignment-09	`main.c`, `README.md`, `output.png`
Assignment-10	`main.c`, `README.md`, `output.png`

README.md Template

Section	Description
Title	Assignment Name
Objective	Purpose of the assignment
Problem Statement	Brief description of the problem
Algorithm	Step-by-step logic
Time Complexity	Big O notation
Space Complexity	Memory analysis
Sample Input	Test input
Sample Output	Expected output
Learning Outcome	Skills acquired

Evaluation Rubric

Evaluation Criteria	Marks
Program Logic & Correctness	30
Code Quality & Comments	15
Functions & Modular Design	10
Arrays / Pointers / Structures Usage	15
Searching & Sorting Implementation	10
Complexity Analysis	5
README Documentation	5
GitHub Repository Structure	5
Commit History & Version Control	5

Total	**100**